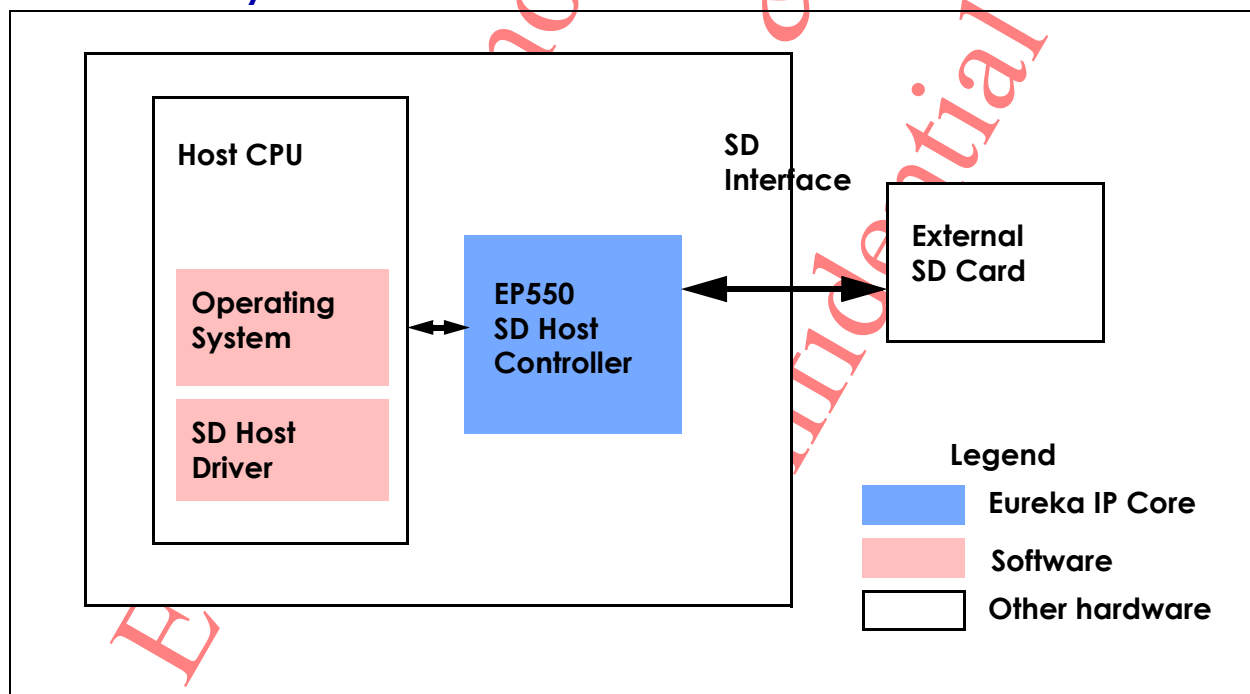




1. Introduction

This application note provides a simplified view of the hardware and software components of an SD system and their relationship to Eureka IP cores. The purpose of this document is to serve as a guide to customer to decide which type of software and external hardware is needed to complete their system development.

2. SD Host System



The above diagram shows a typical application of the EP550 SD Host Controller. The Host Controller connects the host CPU to the external SD card through the SD Interface. Because the EP550 SD Host Controller is compliant to the SD 1.1 and 2.0 specification (MMC 4.2 if the MMC option is added), it is compatible with all standard external SD and/or MMC cards.

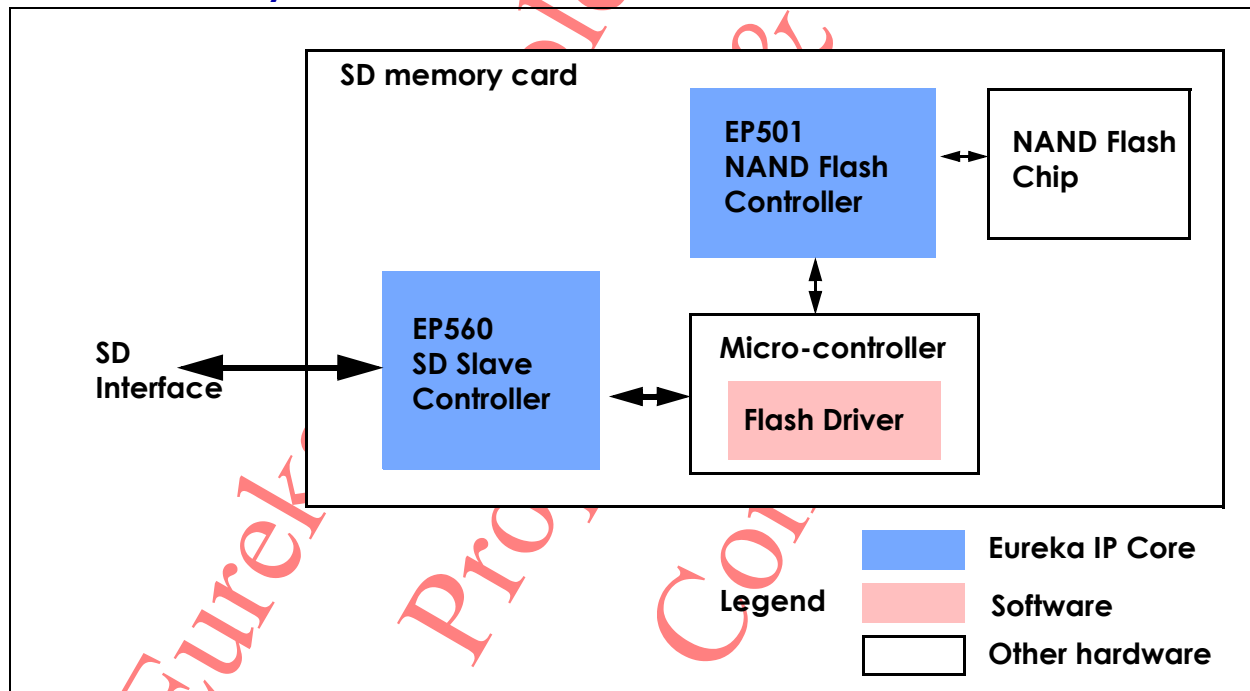
The host CPU executes the operating system (RTOS in most embedded applications) and the operating system should have built-in support for SD slots, including file system for SD memory. In case the operating system does not have built-in support of SD slots, an SD host driver is needed. The SD driver is not specific to the EP550 SD Host Controller. The EP550 implements the standard SD1.1 and 2.0 register set, as long as the SD driver conforms to the SD1.1 or SD2.0 speci-



cation, it will operate with the EP550.

In case it is needed, Eureka can provide SD host driver and related software through our software partners. Because the EP550 SD Host Controller conforms to the SD specification, it works well with any standard SD system software.

3. SD Memory Card



The above diagram shows a typical application of the EP560 SD Slave Controller for implementing SD memory card. The EP560 Slave Controller connects the card to the SD slot of the host system. Requests from the host system for memory access are forwarded by the EP560 to the on-chip micro-controller. The main task of the micro-controller is to execute the flash driver software which includes algorithm such as wear leveling, bad block management, physical/logic address translation and the block replacement.

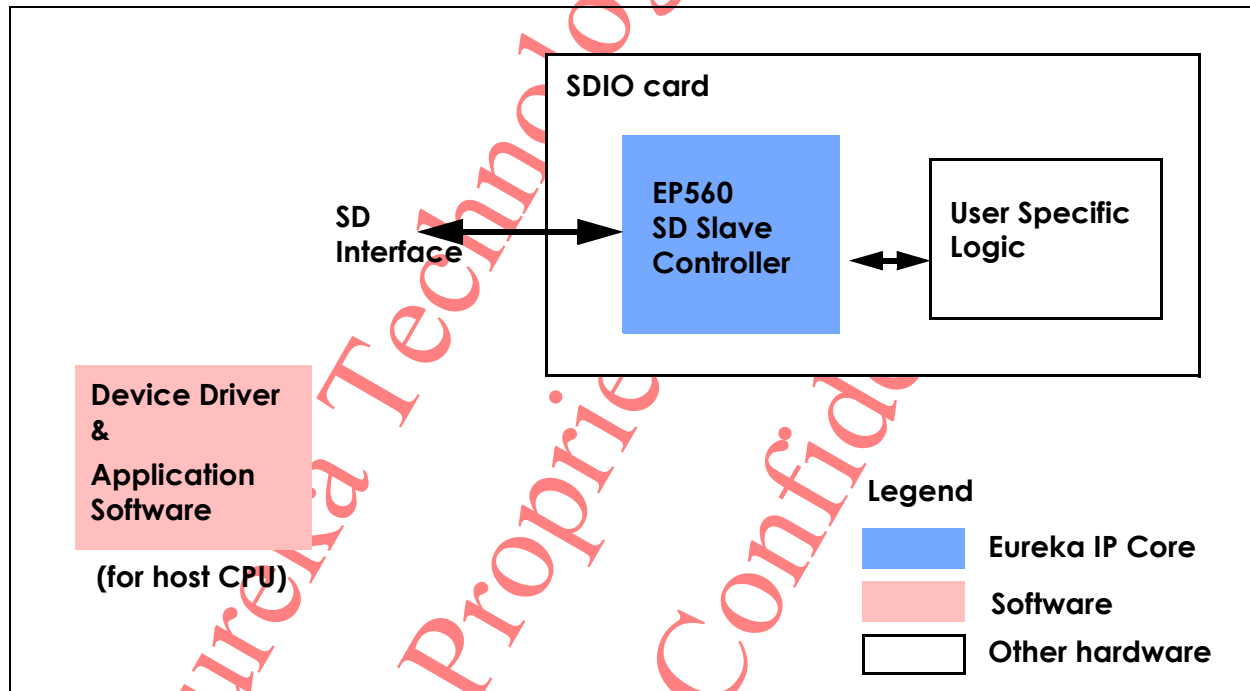
The micro-controller then access a NAND Flash Controller such as the Eureka EP501 IP core which performs the physical layer access to the NAND flash memory chips for data storage.

The flash driver software is typically a low level software runs in the micro-controller. It is a standard software package provide by system software vendors and is independent of either the EP560 or the EP501 IP core. Eureka can provide flash driver software through our software partners even though it operates independently from our EP560 SD Slave Controller.



Because the EP560 conforms to the SD/MMC standard, this memory card can be connected to any host system that accepts standard SD or MMC memory cards. No special software is needed for the host system.

4. SDIO Card



The above diagram shows a typical application of the EP560 SD Slave Controller for implementing SDIO card. The EP560 Slave Controller connects the card to the SD slot of the host system. Requests from the host system for device access are forwarded by the EP560 to the on-chip user specific logic. The design user logic varies widely and its function depends entirely on the specific application of the SDIO card.

Like all hardware peripheral cards, application software and device drivers should be installed in the host system (where the card is plugged-in) in order for the card to operate correctly. The application software and device drivers are specific to the user logic and is independent of the EP560 SD Slave Controller. To the host system software, the EP560 is transparent and only the user specific logic in the card is visible.

Because the application software and device driver closely couple with the user logic, they should be custom made for the user's card function. Eureka can provide reference to software developer for carrying such task through our software partners.



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